

$BRG, DRB, DRG, BRD \vdash B \Vdash A, G \Vdash A, BRG$	identity	$DRB, DRG, BRD \vdash DRB, B \Vdash A, G \Vdash A, BRG$	identity	$DRB, DRG, BRD \vdash DRG, B \Vdash A, G \Vdash A, BRG$	identity
		$DRB, DRG, BRD \vdash (DRB) \wedge (DRG), B \Vdash A, G \Vdash A, BRG$	implyL		andR
$((DRB) \wedge (DRG)) \rightarrow (BRG), DRB, DRG, BRD \vdash B \Vdash A, G \Vdash A, BRG$	forallL	$\forall v.((DRB) \wedge (DRG)) \rightarrow (BRv), DRB, DRG, BRD \vdash B \Vdash A, G \Vdash A, BRG$	forallL	$\forall u.\forall v.((DRu) \wedge (DRv)) \rightarrow (uRv), DRB, DRG, BRD \vdash B \Vdash A, G \Vdash A, BRG$	forallL
$\forall w.\forall u.\forall v.((wRu) \wedge (wRv)) \rightarrow (uRv), DRB, DRG, BRD \vdash B \Vdash A, G \Vdash A, BRG$	forallL	$BRD, \forall w.\forall u.\forall v.((wRu) \wedge (wRv)) \rightarrow (uRv), DRB, DRG \vdash B \Vdash A, G \Vdash A, BRG$	shuffleL	$DRG, BRD, \forall w.\forall u.\forall v.((wRu) \wedge (wRv)) \rightarrow (uRv), DRB \vdash B \Vdash A, G \Vdash A, BRG$	shuffleL
$DRB, DRG, BRD, \forall w.\forall u.\forall v.((wRu) \wedge (wRv)) \rightarrow (uRv) \vdash B \Vdash A, G \Vdash A, BRG$	shuffleL	$DRG, BRD, \forall w.\forall u.\forall v.((wRu) \wedge (wRv)) \rightarrow (uRv) \vdash (DRB) \rightarrow (B \Vdash A), G \Vdash A, BRG$	implyR	$DRG, BRD, \forall w.\forall u.\forall v.((wRu) \wedge (wRv)) \rightarrow (uRv) \vdash \exists E.(DRE) \rightarrow (E \Vdash A), G \Vdash A, BRG$	existsR
$DRG, BRD, \forall w.\forall u.\forall v.((wRu) \wedge (wRv)) \rightarrow (uRv) \vdash G \Vdash A, \exists E.(DRE) \rightarrow (E \Vdash A), BRG$	exchangeR	$BRD, \forall w.\forall u.\forall v.((wRu) \wedge (wRv)) \rightarrow (uRv) \vdash (DRG) \rightarrow (G \Vdash A), \exists E.(DRE) \rightarrow (E \Vdash A), BRG$	implyR	$BRD, \forall w.\forall u.\forall v.((wRu) \wedge (wRv)) \rightarrow (uRv) \vdash \exists E.(DRE) \rightarrow (E \Vdash A), BRG$	existsR
$BRD, \forall w.\forall u.\forall v.((wRu) \wedge (wRv)) \rightarrow (uRv) \vdash BRG, \exists E.(DRE) \rightarrow (E \Vdash A)$	exchangeR		contractionR		
$(BRG) \rightarrow (G \Vdash A), BRD, \forall w.\forall u.\forall v.((wRu) \wedge (wRv)) \rightarrow (uRv) \vdash \exists E.(DRE) \rightarrow (E \Vdash A)$	existsL	$\exists F.(BRF) \rightarrow (F \Vdash A), BRD, \forall w.\forall u.\forall v.((wRu) \wedge (wRv)) \rightarrow (uRv) \vdash \exists E.(DRE) \rightarrow (E \Vdash A)$	mDiamondL	$B \Vdash \Diamond A, BRD, \forall w.\forall u.\forall v.((wRu) \wedge (wRv)) \rightarrow (uRv) \vdash \exists E.(DRE) \rightarrow (E \Vdash A)$	exchangeL
$BRD, B \Vdash \Diamond A, \forall w.\forall u.\forall v.((wRu) \wedge (wRv)) \rightarrow (uRv) \vdash \exists E.(DRE) \rightarrow (E \Vdash A)$	mDiamondR	$BRD, B \Vdash \Diamond A, \forall w.\forall u.\forall v.((wRu) \wedge (wRv)) \rightarrow (uRv) \vdash D \Vdash \Diamond A$	implyR	$B \Vdash \Diamond A, \forall w.\forall u.\forall v.((wRu) \wedge (wRv)) \rightarrow (uRv) \vdash (BRD) \rightarrow (D \Vdash \Diamond A)$	forallR
$B \Vdash \Diamond A, \forall w.\forall u.\forall v.((wRu) \wedge (wRv)) \rightarrow (uRv) \vdash \forall C.(BRC) \rightarrow (C \Vdash \Diamond A)$	mBoxR	$B \Vdash \Diamond A, \forall w.\forall u.\forall v.((wRu) \wedge (wRv)) \rightarrow (uRv) \vdash B \Vdash \Box(\Diamond A)$	implyR	$\forall w.\forall u.\forall v.((wRu) \wedge (wRv)) \rightarrow (uRv) \vdash (B \Vdash \Diamond A) \rightarrow (B \Vdash \Box(\Diamond A))$	mImplyR
$\forall w.\forall u.\forall v.((wRu) \wedge (wRv)) \rightarrow (uRv) \vdash B \Vdash (\Diamond A) \rightarrow (\Box(\Diamond A))$	forallR	$\forall w.\forall u.\forall v.((wRu) \wedge (wRv)) \rightarrow (uRv) \vdash \forall t.t \Vdash (\Diamond A) \rightarrow (\Box(\Diamond A))$	implyR		